BT GATT Protocol Specification - BQ Device

Overview

This document defines the Bluetooth Low Energy (BLE) GATT services and characteristics for the BQ pill dispenser device. Uses the BLEOTA library for OTA firmware updates.

Custom services use 128-bit UUIDs with the base format: `XXXXXXXX-0000-1000-8000-00805F9B34FB`.

Service Summary

Service Name	Service UUID	Description
Device Information (DIS)	`0x180A`	Device identification (standard BLE DIS)
BLE OTA	`00008018-0000-1000-8000-00805f9b34fb`	Firmware update via BLEOTA protocol
Time Sync	`0x1802`	Device clock synchronization
Terminal Command	`0x1803`	Debug/diagnostic terminal
Device Sensors	`0x1804`	Sensor status monitoring
Device Control	`0x1805`	Movement and control commands
Extract Pill	`0x1806`	Pill extraction operations
Sub Drive	`0x1807`	Sub drive information
Notifications	`0x1808`	Event notifications
Offline Sync	`0x1809`	Offline data synchronization
Battery	`0x180F`	Battery level (standard BLE)
Sound File	`0x180B`	Audio file management

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1. Device Information Service (DIS)

Service UUID: `0x180A` (Standard BLE DIS)

Provides device identification and status information. Uses the standard Bluetooth Device Information Service combined with BLEOTA library methods.

Characteristics

Characteristic	UUID	Properties	Description	API Method
Model Number	`0x2A24`	Read	Device model string	`setModel()`
Serial Number	`0x2A25`	Read	Device serial number	`setSerialNumber()`
Firmware Revision	`0x2A26`	Read	Firmware version string	`setFWVersion()`
Hardware Revision	`0x2A27`	Read	Hardware revision string	`setHWVersion()`
Manufacturer Name	`0x2A29`	Read	Manufacturer identifier	`setManufactuer()`

Battery Service

Battery level is provided via a separate standard Battery Service (`0x180F`):

Characteristic	UUID	Properties	Description
Battery Level	`0x2A19`	Read, Notify	Current battery percentage (0-100)

Battery Level Format

```
Byte 0: Battery percentage (0x00 - 0x64)
```

Firmware Version Format

```
Bytes 0-N: Major.Minor.Patch as ASCII string (null-terminated)
Example: "1.2.3"
```

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2. BLE OTA Service (BLEOTA)

Service UUID: `00008018-0000-1000-8000-00805f9b34fb`

Reference: github.com/gb88/BLEOTA

Handles over-the-air (OTA) firmware and SPIFFS filesystem updates using the BLEOTA protocol. Supports zlib compression (~30% faster transfers) and RSA-2048 signature verification.

Characteristics

Characteristic	UUID	Properties	Description
Firmware Receive	`00008020-0000-1000-8000-00805f9b34fb`	Write, WriteNR, Notify	Firmware data reception with ACK
Command	`00008022-0000-1000-8000-00805f9b34fb`	Write, WriteNR, Notify	OTA commands and responses

Command Packet Format

```
Bytes 0-1: Command ID (uint16, little-endian)
Bytes 2-17: Payload (16 bytes, zero-padded if unused)
Bytes 18-19: CRC16 checksum

Total: 20 bytes
```

Command Codes

Command	ID	Payload	Description
Start Flash OTA	`0x0001`	Bytes 2-5: Firmware size (uint32)	Initialize firmware update
Start SPIFFS OTA	`0x0004`	Bytes 2-5: Filesystem size (uint32)	Initialize SPIFFS update
Stop OTA	`0x0002`	None	Abort current update
Response	`0x0003`	Status code	Device response to commands

Command Response Status Codes

Status	Value	Description
Accept	`0x0000`	Command accepted, OTA started
Reject	`0x0001`	Command rejected
Signature Error	`0x0003`	RSA signature verification failed

Firmware Data Packet Format (App → Device)

```
Bytes 0-1: Sector Index (uint16, little-endian)
Byte 2: Packet Sequence (0x00-0xFE for data, 0xFF for final packet)
Bytes 3-N: Payload data (MTU - 4 bytes, typically up to 243 bytes)

Note: When Packet_Seq = 0xFF, the final 2 bytes contain CRC16 of the sector
```

Firmware ACK Format (Device → App via Notify)

```
Bytes 0-1: Sector Index (uint16, little-endian)
Bytes 2-3: ACK Status (uint16, little-endian)
Bytes 4-5: CRC16 checksum

Total: 6 bytes
```

ACK Status Codes

Status	Value	Description
Success	`0x0000`	Sector received successfully
CRC Error	`0x0001`	CRC verification failed
Sector Mismatch	`0x0002`	Unexpected sector index
Length Error	`0x0003`	Invalid payload length
OTA Start Failure	`0x0005`	Failed to initialize OTA

OTA Update Flow

```
┌─────────────────────────────────────────────────────────────────────┐
│                         OTA UPDATE FLOW                             │
├─────────────────────────────────────────────────────────────────────┤
│                                                                     │
│  APP                                      DEVICE                    │
│    │                                         │                      │
│    │─── Write Command: Start OTA ──────────>│                      │
│    │    (0x0001, size=firmware_size)         │                      │
│    │                                         │                      │
│    │<── Notify Response: Accept ────────────│                      │
│    │    (0x0003, status=0x0000)              │                      │
│    │                                         │                      │
│    │    ┌─── FOR EACH SECTOR (4KB) ──────┐  │                      │
│    │    │                                │  │                      │
│    │    │  Write FW Data: Packet 0       │  │                      │
```

```
│    │─────│──(sector_idx, seq=0, data)───────│─>                    │
│    │     │                                  │ │                    │
│    │     │  Write FW Data: Packet 1...N-1   │ │                    │
│    │─────│──(sector_idx, seq=1..N-1, data)─│─>                    │
│    │     │                                  │ │                    │
│    │     │  Write FW Data: Final Packet     │ │                    │
│    │─────│──(sector_idx, seq=0xFF, CRC16)──│─>                    │
│    │     │                                  │ │                    │
│    │<────│── Notify ACK ────────────────────│─│                    │
│    │     │   (sector_idx, status, CRC)      │ │                    │
│    │     │                                  │ │                    │
│    │     └──────────────────────────────────┘ │                    │
│    │                                          │                    │
│    │    [All sectors transmitted]             │                    │
│    │                                          │                    │
│    │<── Device resets (500ms delay) ──────────│                    │
│    │                                          │                    │
└──────────────────────────────────────────────────────────────────┘
```

Sector Transmission Details

Parameter	Value
Sector Size	4096 bytes (4KB)
Packet Size	MTU - 4 bytes (typically 243 bytes with MTU=247)
Packets per Sector	~17 packets for 4KB sector
Sequence Numbers	0x00 to 0xFE for data, 0xFF signals final packet

Security Features

Feature	Description
CRC16	Per-sector integrity check
RSA-2048	Firmware signature verification using SHA256 hash
Compression	Optional zlib compression for faster transfer (~30% improvement)

Post-Update Behavior

After successful OTA completion:

1. Device validates firmware signature (if enabled)

2. Device waits 500ms (configurable)
3. Device resets and boots new firmware

Error Recovery

Scenario	Recovery
CRC Error	Retransmit current sector
Timeout	Restart from last ACK'd sector
Signature Fail	OTA rejected, device unchanged
Connection Lost	Resume from last successful sector

Example: Calculate Transfer Time

```
Firmware size: 1MB (1,048,576 bytes)
Sector size: 4KB
Total sectors: 256
Packets per sector: ~17
Connection interval: 30ms
Packets per interval: 6 (conservative)

Time per sector: 17 / 6 * 30ms ≈ 85ms
Total time: 256 * 85ms ≈ 22 seconds

With zlib compression (~30% reduction):
Compressed size: ~734KB
Total time: ≈ 15 seconds
```

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3. Time Sync Service

Service UUID: `0x1802`

Synchronizes device clock with mobile app/server time.

Characteristics

Characteristic	UUID	Properties	Description
Current Time	`0x2C01`	Read, Write	Device current time
Time Zone	`0x2C02`	Read, Write	Timezone offset

Characteristic	UUID	Properties	Description
DST Offset	`0x2C03`	Read, Write	Daylight saving offset
Last Sync	`0x2C04`	Read	Timestamp of last sync

Current Time Format

```
Bytes 0-3: Unix timestamp (uint32, little-endian, seconds since 1970)
Bytes 4-5: Milliseconds (uint16, little-endian)
```

Time Zone Format

```
Byte 0: Timezone offset in 15-minute increments (int8)
        Example: UTC+2 = 0x08 (8 * 15 = 120 minutes)
```

DST Offset Format

```
Byte 0: DST offset in minutes (uint8)
        0x00 = No DST
        0x3C = 60 minutes DST
```

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4. Terminal Command Service

Service UUID: `0x1803`

Provides debug and diagnostic terminal access.

Characteristics

Characteristic	UUID	Properties	Description
TX (Device→App)	`0x2D01`	Notify	Terminal output from device
RX (App→Device)	`0x2D02`	Write	Terminal input to device
Command Status	`0x2D03`	Read, Notify	Command execution status

Command Format

```
Bytes 0-N: ASCII command string (null-terminated, max 200 bytes)
```

Response Format

```
Byte 0: Response type
  0x00 = Output line
  0x01 = Command complete (success)
  0x02 = Command complete (error)
  0x03 = Command in progress
Bytes 1-N: ASCII response data
```

Common Terminal Commands

Command	Description
`help`	List available commands
`status`	Device status summary
`reset`	Soft reset device
`factory`	Factory reset (requires confirmation)
`log [level]`	Set/get log level
`diag`	Run diagnostics

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5. Device Sensors Service

Service UUID: `0x1804`

Monitors device sensor status and readings.

Characteristics

Characteristic	UUID	Properties	Description
Sensor Status	`0x2E01`	Read, Notify	Combined sensor status bitmap
Temperature	`0x2E02`	Read, Notify	Internal temperature
Humidity	`0x2E03`	Read, Notify	Internal humidity
Motor Status	`0x2E04`	Read, Notify	Motor position and status
Lid Status	`0x2E05`	Read, Notify	Lid open/closed status
Pill Detection	`0x2E06`	Read, Notify	Pill presence sensors

Sensor Status Bitmap

```
Byte 0: Status flags
  Bit 0: Temperature sensor OK
  Bit 1: Humidity sensor OK
  Bit 2: Motor encoder OK
  Bit 3: Lid sensor OK
  Bit 4: Pill sensors OK
  Bit 5: Battery sensor OK
  Bit 6-7: Reserved
Byte 1: Error flags
  Bit 0: Temperature out of range
  Bit 1: Humidity out of range
  Bit 2: Motor fault
  Bit 3: Lid jam detected
  Bit 4: Pill sensor fault
  Bit 5-7: Reserved
```

Temperature Format

```
Bytes 0-1: Temperature in 0.01°C (int16, little-endian)
           Example: 2350 = 23.50°C
```

Motor Status Format

```
Byte 0: Current position (0-11 for 12 slots)
Byte 1: Target position
Byte 2: Motor state
  0x00 = Idle
  0x01 = Moving
  0x02 = Homing
  0x03 = Error
Byte 3: Error code (if state = 0x03)
```

Pill Detection Format

```
Bytes 0-1: Pill presence bitmap (uint16)
           Bit N = 1 if pill present in slot N
           Example: 0x0FFF = all 12 slots have pills
```

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6. Device Control Service

Service UUID: `0x1805`

Controls device movement and operations.

Characteristics

Characteristic	UUID	Properties	Description
Control Command	`0x2F01`	Write, Indicate	Issue control commands
Control Status	`0x2F02`	Read, Notify	Current operation status
Position	`0x2F03`	Read, Write, Notify	Current/target position

Control Commands

Command	Value	Parameters	Description
Home	`0x01`	None	Return to home position
Rotate	`0x02`	Position (1 byte)	Rotate to position 0-11
Extract	`0x03`	Position (1 byte)	Extract pill at position
Stop	`0x04`	None	Emergency stop
Calibrate	`0x05`	None	Run calibration sequence

Command Request Format

```
Byte 0: Command code
Bytes 1-N: Command parameters (varies by command)
```

Command Response Format (Indicate)

```
Byte 0: Original command code
Byte 1: Result code
  0x00 = Success
  0x01 = Invalid command
  0x02 = Invalid parameter
  0x03 = Busy
  0x04 = Hardware error
  0x05 = Timeout
Bytes 2-N: Result data (varies by command)
```

Control Status Format

```
Byte 0: Device state
  0x00 = Idle
  0x01 = Executing command
  0x02 = Error state
  0x03 = Calibrating
Byte 1: Last command code
Byte 2: Last command result
Byte 3: Current position (0-11)
```

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7. Extract Pill Service

Service UUID: `0x1806`

Manages pill extraction operations.

Characteristics

Characteristic	UUID	Properties	Description
Extract Command	`0x3001`	Write, Indicate	Initiate extraction
Extract Status	`0x3002`	Read, Notify	Extraction operation status
Extract History	`0x3003`	Read	Recent extraction records
Schedule	`0x3004`	Read, Write	Scheduled extractions

Extract Command Format

```
Byte 0: Command type
  0x01 = Extract from position
  0x02 = Extract next scheduled
  0x03 = Cancel extraction
Byte 1: Position (0-11, for type 0x01)
Byte 2: Confirmation flag (0x5A = confirmed)
```

Extract Status Format

```
Byte 0: Extraction state
  0x00 = Idle
```

```
  0x01 = Rotating to position
  0x02 = Extracting
  0x03 = Verifying
  0x04 = Complete (success)
  0x05 = Complete (failed)
Byte 1: Current position
Byte 2: Target position
Byte 3: Error code (if failed)
  0x00 = No error
  0x01 = No pill detected
  0x02 = Jam detected
  0x03 = Motor error
  0x04 = Timeout
```

Extract History Format

```
Each record (8 bytes):
  Bytes 0-3: Timestamp (uint32, Unix time)
  Byte 4: Position extracted
  Byte 5: Result (0x00 = success, other = error code)
  Bytes 6-7: Reserved

Max 10 records returned
```

Schedule Format

```
Each schedule entry (6 bytes):
  Byte 0: Day mask (bit 0 = Sunday, bit 6 = Saturday)
  Byte 1: Hour (0-23)
  Byte 2: Minute (0-59)
  Byte 3: Position (0-11)
  Byte 4: Enabled flag (0x00 = disabled, 0x01 = enabled)
  Byte 5: Reserved

Max 14 schedule entries
```

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8. Sub Drive Service

Service UUID: `0x1807`

Provides sub drive (cartridge) information.

Characteristics

Characteristic	UUID	Properties	Description
Sub Drive Info	`0x3101`	Read, Notify	Cartridge identification
Sub Drive Status	`0x3102`	Read, Notify	Cartridge status
Pill Count	`0x3103`	Read, Notify	Pills remaining per slot
Medication Info	`0x3104`	Read	Medication metadata

Sub Drive Info Format

```
Bytes 0-15: Cartridge UUID (128-bit)
Bytes 16-19: Manufacturing date (uint32, Unix time)
Bytes 20-23: Expiration date (uint32, Unix time)
Byte 24: Total slots (typically 12)
Byte 25: Cartridge type
  0x01 = Standard
  0x02 = Extended
  0x03 = Custom
```

Sub Drive Status Format

```
Byte 0: Status
  0x00 = Not inserted
  0x01 = Inserted, valid
  0x02 = Inserted, expired
  0x03 = Inserted, error
Byte 1: Error code (if status = 0x03)
Bytes 2-3: Total extractions (uint16)
Bytes 4-5: Successful extractions (uint16)
```

Pill Count Format

```
Bytes 0-11: Pill count per slot (1 byte each, 0-255)
            Slot 0 at byte 0, slot 11 at byte 11
Byte 12: Total pills remaining
```

Medication Info Format

```
Bytes 0-31: Medication name (UTF-8, null-terminated)
Bytes 32-47: NDC code (ASCII, null-terminated)
Bytes 48-51: Dosage (uint32, micrograms)
Byte 52: Dosage unit
  0x01 = mg
  0x02 = mcg
  0x03 = IU
```

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9. Notifications Service

Service UUID: `0x1808`

Handles event notifications and alerts.

Characteristics

Characteristic	UUID	Properties	Description
Notification	`0x3201`	Notify	Push notifications to app
Notification Config	`0x3202`	Read, Write	Notification settings
Alert Status	`0x3203`	Read, Notify	Active alert status
Acknowledge	`0x3204`	Write	Acknowledge alerts

Notification Format

```
Byte 0: Notification type
  0x01 = Scheduled extraction reminder
  0x02 = Extraction complete
  0x03 = Extraction failed
  0x04 = Low battery
  0x05 = Cartridge empty
  0x06 = Cartridge expiring
  0x07 = Device error
  0x08 = Connectivity lost
Bytes 1-4: Timestamp (uint32, Unix time)
Byte 5: Priority (0x01 = low, 0x02 = normal, 0x03 = high)
Bytes 6-N: Payload (varies by type, max 100 bytes)
```

Notification Config Format

```
Byte 0: Enabled notifications bitmap
  Bit 0: Scheduled reminders
  Bit 1: Extraction events
  Bit 2: Battery alerts
  Bit 3: Cartridge alerts
  Bit 4: Error alerts
  Bit 5-7: Reserved
Byte 1: Reminder lead time (minutes before scheduled)
Byte 2: Low battery threshold (percentage)
Byte 3: Quiet hours start (hour, 0-23)
Byte 4: Quiet hours end (hour, 0-23)
```

Alert Status Format

```
Byte 0: Active alerts bitmap
  Bit 0: Low battery
  Bit 1: Empty slot
  Bit 2: Expired cartridge
  Bit 3: Motor error
  Bit 4: Sensor error
  Bit 5: Missed dose
  Bit 6-7: Reserved
Bytes 1-4: Oldest alert timestamp
```

Acknowledge Format

```
Byte 0: Alert type to acknowledge (from alert bitmap)
        0xFF = Acknowledge all
```

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10. Offline Sync Service

Service UUID: `0x1809`

Synchronizes offline data (sub drive logs, telemetry).

Characteristics

Characteristic	UUID	Properties	Description
Sync Control	`0x3301`	Write, Indicate	Control sync operations

Characteristic	UUID	Properties	Description
Sync Data	`0x3302`	Notify	Streamed sync data
Sync Status	`0x3303`	Read, Notify	Sync progress status
Last Sync Time	`0x3304`	Read	Last successful sync timestamp

Sync Control Commands

Command	Value	Parameters	Description
Start Sync	`0x01`	Start timestamp (4 bytes)	Begin sync from timestamp
Pause Sync	`0x02`	None	Pause current sync
Resume Sync	`0x03`	None	Resume paused sync
Abort Sync	`0x04`	None	Cancel sync operation
Mark Complete	`0x05`	End timestamp (4 bytes)	Mark sync complete

Sync Data Format

```
Byte 0: Record type
  0x01 = Extraction event
  0x02 = Error event
  0x03 = Telemetry sample
  0x04 = State change
Bytes 1-4: Timestamp (uint32, Unix time)
Byte 5: Sequence number (for ordering)
Bytes 6-N: Record payload (varies by type)
```

Extraction Event Payload

```
Byte 0: Position
Byte 1: Result (0x00 = success, other = error)
Byte 2: Duration (seconds)
Byte 3: Battery level at time
```

Telemetry Sample Payload

```
Bytes 0-1: Temperature (int16, 0.01°C)
Bytes 2-3: Humidity (uint16, 0.01%)
```

```
Byte 4: Battery level
Byte 5: Motor status
```

Sync Status Format

```
Byte 0: Sync state
  0x00 = Idle
  0x01 = Syncing
  0x02 = Paused
  0x03 = Error
Bytes 1-2: Records pending (uint16)
Bytes 3-4: Records synced (uint16)
Byte 5: Progress percentage (0-100)
```

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11. Sound File Service

Service UUID: `0x180B`

Manages audio files for alerts and notifications.

Characteristics

Characteristic	UUID	Properties	Description
Sound Control	`0x3401`	Write, Indicate	Sound playback control
Sound Upload	`0x3402`	Write Without Response	Upload sound data
Sound List	`0x3403`	Read	List available sounds
Sound Config	`0x3404`	Read, Write	Sound configuration
Volume	`0x3405`	Read, Write	Master volume level

Sound Control Commands

Command	Value	Parameters	Description
Play	`0x01`	Sound ID (1 byte)	Play sound by ID
Stop	`0x02`	None	Stop current playback
Upload Start	`0x03`	Sound ID, Size (4 bytes)	Begin upload
Upload End	`0x04`	Checksum (4 bytes)	Complete upload

Command	Value	Parameters	Description
Delete	`0x05`	Sound ID (1 byte)	Delete sound file

Sound Upload Format

```
Bytes 0-1: Packet sequence (uint16)
Bytes 2-N: PCM audio data (8kHz, 8-bit mono, max 244 bytes)
```

Sound List Format

```
Each entry (20 bytes):
  Byte 0: Sound ID
  Bytes 1-16: Sound name (UTF-8, null-terminated)
  Bytes 17-18: Duration (uint16, milliseconds)
  Byte 19: Type
    0x01 = System
    0x02 = Alert
    0x03 = Custom
```

Sound Config Format

```
Byte 0: Reminder sound ID
Byte 1: Success sound ID
Byte 2: Error sound ID
Byte 3: Low battery sound ID
Byte 4: Extraction sound ID
```

Volume Format

```
Byte 0: Volume level (0-100)
Byte 1: Mute flag (0x00 = not muted, 0x01 = muted)
```

Error Codes Reference

Global Error Codes

Code	Name	Description
`0x00`	SUCCESS	Operation completed successfully
`0x01`	ERR_INVALID_PARAM	Invalid parameter value
`0x02`	ERR_INVALID_STATE	Operation not valid in current state
`0x03`	ERR_BUSY	Device busy with another operation
`0x04`	ERR_TIMEOUT	Operation timed out
`0x05`	ERR_NOT_FOUND	Requested resource not found
`0x06`	ERR_NO_MEMORY	Insufficient memory
`0x07`	ERR_HARDWARE	Hardware fault detected
`0x08`	ERR_COMM	Communication error
`0x09`	ERR_AUTH	Authentication required/failed
`0x0A`	ERR_NOT_SUPPORTED	Feature not supported
`0xFF`	ERR_UNKNOWN	Unknown error

Connection Parameters

Recommended BLE Parameters

Parameter	Value	Description
Connection Interval	30-50 ms	Balance between latency and power
Slave Latency	0-4	Allow skipping connection events
Supervision Timeout	4000 ms	Connection timeout
MTU	247 bytes	Maximum transmission unit

Advertising Parameters

Parameter	Value
Advertising Interval	100-200 ms (fast), 1000-2000 ms (slow)
Advertising Type	Connectable, Scannable
TX Power	0 dBm

Service Discovery

The device advertises the following service UUIDs:

- Device Information Service (`0x180A`)
- BLE OTA Service (`00008018-0000-1000-8000-00805f9b34fb`)
- Device Control Service (`0x1805`)

Full service list available after connection via GATT discovery.

Security

Pairing Requirements

- **Bonding:** Required
- **MITM Protection:** Required
- **Secure Connections:** Supported (LE Secure Connections)
- **IO Capability:** Display + Yes/No

Encryption

All characteristics except Device Information Service (read-only info) require an encrypted connection.

Authentication

Sensitive operations (FW Update, Factory Reset, Extract) require:

1. Encrypted connection
2. Bonded device
3. Optional: PIN confirmation for critical operations

Version History

Version	Date	Changes
1.0	TBD	Initial specification